

Experiment 5

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/types.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    int pipefd[2];
```

```
    pid_t pid1, pid2;
```

```
    // Create pipe
```

```
    if (pipe(pipefd) == -1)
```

```
    {
```

```
        perror("pipe failed");
```

```
        exit(1);
```

```
    }
```

```
    // Create first child process for ls -l
```

```
    pid1 = fork();
```

```
if (pid1 == -1)
{
    perror("fork failed");
    exit(1);
}

if (pid1 == 0)
{
    // Child 1: ls -l

    close(pipefd[0]);          // Close read end
    dup2(pipefd[1], STDOUT_FILENO); // Redirect stdout to pipe
    close(pipefd[1]);

    execlp("ls", "ls", "-l", NULL);

    perror("execlp ls failed");
    exit(1);
}

// Create second child process for grep ".c"
pid2 = fork();

if (pid2 == -1)
```

```
{
    perror("fork failed");
    exit(1);
}

if (pid2 == 0)
{
    // Child 2: grep ".c"

    close(pipefd[1]);      // Close write end
    dup2(pipefd[0], STDIN_FILENO); // Redirect stdin from pipe
    close(pipefd[0]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp grep failed");
    exit(1);
}

// Parent process
close(pipefd[0]);
close(pipefd[1]);

// Wait for both children
```

```
waitpid(pid1, NULL, 0);
```

```
waitpid(pid2, NULL, 0);
```

```
return 0;
```

```
}
```

Output:

```
ashwithreddy_earla@ASHWITHREDDYPC:~$ nano Experiment5.c
ashwithreddy_earla@ASHWITHREDDYPC:~$ gcc Experiment5.c -o Experiment5
ashwithreddy_earla@ASHWITHREDDYPC:~$ ./Experiment5
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 779 Aug 25 03:26 Experiment2.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1353 Aug 25 03:33 Experiment5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 11 03:14 OSSP.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16120 Aug 23 06:02 close
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 489 Aug 23 06:01 close.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 30 Aug 25 03:22 copy.txt
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 15992 Aug 11 03:43 exec
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 199 Aug 11 03:43 exec.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 791 Aug 25 03:04 expl.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 99 Aug 3 05:56 expl.program1.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 207 Aug 3 05:59 expl.program3.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 419 Jul 31 09:58 expl.program5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 127 Aug 3 05:44 expl.program8.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1177 Aug 25 03:22 experiment2.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1019 Aug 23 15:08 experiment3.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1500 Aug 23 15:10 experiment4.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 204 Aug 3 05:59 experiment5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 112 Aug 3 05:39 experiment6.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 209 Aug 3 05:40 experiment7.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 452 Aug 11 03:44 fork.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16352 Aug 11 03:50 fork_exec
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 609 Aug 11 03:49 fork_exec.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1118 Jul 28 04:07 gcc
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 510 Aug 18 03:32 input.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 443 Aug 23 05:51 open.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16864 Aug 24 05:22 produce
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1868 Aug 24 05:21 produce.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 693 Aug 23 05:54 read.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 908 Jul 31 10:14 sample.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1 Aug 3 06:09 sample.c.save
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 774 Aug 24 15:09 skill.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 48 Aug 25 03:28 source.txt
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 637 Aug 23 05:56 write.c
ashwithreddy_earla@ASHWITHREDDYPC:~$
ashwithreddy_earla@ASHWITHREDDYPC:~$
ashwithreddy_earla@ASHWITHREDDYPC:~$
ashwithreddy_earla@ASHWITHREDDYPC:~$ ls -l | grep ".c"
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 779 Aug 25 03:26 Experiment2.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1353 Aug 25 03:33 Experiment5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 0 Aug 11 03:14 OSSP.c
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 16120 Aug 23 06:02 close
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 489 Aug 23 06:01 close.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 30 Aug 25 03:22 copy.txt
-rwxr-xr-x 1 ashwithreddy_earla ashwithreddy_earla 15992 Aug 11 03:43 exec
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 199 Aug 11 03:43 exec.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 791 Aug 25 03:04 expl.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 99 Aug 3 05:56 expl.program1.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 207 Aug 3 05:59 expl.program3.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 419 Jul 31 09:58 expl.program5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 127 Aug 3 05:44 expl.program8.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1177 Aug 25 03:22 experiment2.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1019 Aug 23 15:08 experiment3.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 1500 Aug 23 15:10 experiment4.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 204 Aug 3 05:59 experiment5.c
-rw-r--r-- 1 ashwithreddy_earla ashwithreddy_earla 112 Aug 3 05:39 experiment6.c
```